# Notes on MML: Vectors, the Minus-One Trick, Linear Maps, Basis Change, Kernel, and Image

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While reading the linear-algebra chapters of MML, the same objects kept moving between geometric, computational, and structural descriptions. The first example is the word "vector" itself.

## §1.1 Vectors beyond arrows

Vectors are not only geometric arrows. Polynomials form a vector space because they can be added and scaled. Audio signals can likewise be represented by lists of numbers whose addition and scalar multiplication remain valid signals.

Machine-learning embeddings provide another example. Word and sentence vectors are not literal geometric arrows, but they live in vector spaces and obey vector operations; their meaning comes from the learned representation.

The lesson is that a vector is an element of a vector space. Geometry is only one model.

## §1.2 The minus-one trick

The MML notes use a practical trick for reading solutions of a homogeneous system

$$Ax = 0, \qquad A \in \mathbb{R}^{k\times n}.$$

Assume $A$ is in reduced row echelon form and has no zero rows. Pivot columns correspond to leading ones; free columns correspond to variables that can be chosen freely.

The minus-one trick augments the row-reduced matrix into an $n\times n$ matrix by inserting rows with a single $-1$ in the diagonal positions corresponding to free variables. Then the columns containing $-1$ on the diagonal directly give basis vectors for the null space.

The point is not magic. It is a compact way of doing the usual free-variable substitution. Setting one free variable to 1 and the others to 0 gives one kernel basis vector; the inserted $-1$ row records that choice.

## §1.3 Example of the trick

Consider

$$A = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}.$$

The free variables are in columns 2 and 5. Augmenting gives

$$\tilde{A} = \begin{bmatrix} 1 & 3 & 0 & 0 & 3 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 9 \\ 0 & 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}.$$

The columns with diagonal $-1$ give

$$x = \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 0 \\ 9 \\ -4 \\ -1 \end{bmatrix}.$$

This is the same solution one obtains by solving by free variables, but the trick makes the basis easy to read.

## §1.4 Linear mappings

A mapping $\Phi : V \to W$ between vector spaces is linear if

$$\Phi(\lambda x + \psi y) = \lambda\Phi(x) + \psi\Phi(y)$$

for all $x, y \in V$ and scalars $\lambda, \psi$.

Important special cases are:

- an isomorphism is linear and bijective;
- an endomorphism is a linear map $V \to V$;
- an automorphism is a bijective endomorphism;
- the identity map $\mathrm{id}_V : V \to V$ is the identity automorphism.

These notions may be summarized as "same structure," "same domain and codomain," and "self-isomorphism."

## §1.5 Complex numbers as a real vector space

The example

$$\Phi : \mathbb{R}^2 \to \mathbb{C}, \qquad \Phi(x_1, x_2) = x_1 + ix_2$$

is linear over $\mathbb{R}$:

$$\Phi(x + y) = \Phi(x) + \Phi(y), \qquad \Phi(\lambda x) = \lambda\Phi(x).$$

It is also bijective, so it is an isomorphism of real vector spaces.

The note marks this as a homomorphism. The important distinction is that it preserves the vector-space addition and scalar multiplication. It is not claiming that every additional structure is automatically preserved.

## §1.6 Finite-dimensional isomorphism theorem

The theorem quoted from Axler says that finite-dimensional vector spaces $V$ and $W$ are isomorphic iff

$$\dim V = \dim W.$$

This is one of the most important simplifications in finite-dimensional linear algebra: after choosing bases, all $n$-dimensional vector spaces over the same field look like $\mathbb{F}^n$.

But this isomorphism is not canonical unless a basis is chosen. The abstract vector space and its coordinate representation are related but not identical.

## §1.7 Basis change

Let $\Phi : V \to W$ be linear. Suppose $V$ has old basis

$$B = (b_1, \dots, b_n)$$

and new basis

$$\widetilde{B} = (\widetilde{b}_1, \dots, \widetilde{b}_n).$$

Similarly, let $W$ have old basis

$$C = (c_1, \dots, c_m)$$

and new basis

$$\widetilde{C} = (\widetilde{c}_1, \dots, \widetilde{c}_m).$$

If $A_\Phi$ is the matrix of $\Phi$ with respect to $B, C$, and $\widetilde{A}_\Phi$ is the matrix with respect to $\widetilde{B}, \widetilde{C}$, then

$$\widetilde{A}_\Phi = T^{-1} A_\Phi S.$$

Here $S$ converts coordinates from the new basis of $V$ to the old basis of $V$, and $T$ converts coordinates from the new basis of $W$ to the old basis of $W$.

The coordinate-change diagram expresses exactly this procedure: translate the input coordinates into the original basis, apply $A_\Phi$, and then translate the output coordinates into the new output basis.

## §1.8 Equivalence and similarity

Two rectangular matrices $A, \widetilde{A} \in \mathbb{R}^{m\times n}$ are equivalent if there exist invertible matrices $S \in \mathbb{R}^{n\times n}$ and $T \in \mathbb{R}^{m\times m}$ such that

$$\widetilde{A} = T^{-1} A S.$$

This corresponds to changing the basis in both domain and codomain.

Two square matrices $A, \widetilde{A} \in \mathbb{R}^{n\times n}$ are similar if

$$\widetilde{A} = S^{-1} A S.$$

This corresponds to changing basis for a linear map $V \to V$ using the same coordinate change in domain and codomain.

The distinction is:

| Concept | Situation | Invariant meaning |
|---|---|---|
| Equivalence | linear map $V \to W$ | change bases in domain and codomain |
| Similarity | linear map $V \to V$ | same operator in a new basis |

## §1.9 Kernel and image

For a linear map $\Phi : V \to W$, the kernel is

$$\ker \Phi = \{v \in V : \Phi(v) = 0_W\},$$

and the image is

$$\operatorname{Im} \Phi = \{w \in W : \exists v \in V,\ \Phi(v) = w\}.$$

The kernel records what is collapsed to zero. The image records what outputs are reachable.

The source diagram shows the domain on the left, codomain on the right, a yellow region for the kernel, and a blue/yellow region for the image. The note also records the injectivity criterion:

$$\Phi \text{ is injective} \iff \ker \Phi = \{0_V\}.$$

## §1.10 Categorical side remarks

The margin notes compare kernel, image, and cokernel with categorical language. In ordinary linear algebra,

$$\ker f \to A \to B$$

means the kernel is a subobject of the domain. The image is a subobject of the codomain. The cokernel, when defined, is a quotient-like object on the codomain side.

This is a useful preview: linear algebra already contains many category-theoretic patterns, even before they are named abstractly.

## §1.11 The general pattern

This note links computational linear algebra with structural thinking. Row reduction gives the null space; the minus-one trick packages free-variable substitution. Linear maps become matrices only after choosing bases. Basis change explains why the same map can have different matrices. Kernel and image describe what a map destroys and what it reaches. These ideas are also the language behind machine-learning embeddings, where vectors need not be arrows but still live in vector spaces with meaningful linear structure.

## §1.12 Vectors as elements of a linear structure

A vector is not essentially an arrow; it is an element of a vector space. Coordinates appear only after choosing a basis. The same vector has different coordinate columns in different bases.

## §1.13 The minus-one trick as a kernel graph

The phrase “minus-one trick” should be understood carefully. In this note it is not primarily a homogeneous-coordinate trick and it is not an affine shift. It is a compact way of reading the kernel of a row-reduced homogeneous system.

**Proposition 1.13.1** (Kernel graph form of an RREF system)
After possibly reordering variables, suppose a homogeneous system has row-reduced matrix

$$\begin{bmatrix} I_r & A \end{bmatrix}.$$

Write the unknown vector as

$$x = (x_p, x_f),$$

where $x_p \in k^r$ are pivot variables and $x_f \in k^{n-r}$ are free variables. Then the solution space is

$$\ker \begin{bmatrix} I_r & A \end{bmatrix} = \left\{ \begin{pmatrix} -Au \\ u \end{pmatrix} : u \in k^{n-r} \right\}.$$

Thus the kernel is the graph of the linear map

$$u \longmapsto -Au.$$

*Proof.* The equation is

$$x_p + Ax_f = 0.$$

Hence $x_p = -Ax_f$. If we write $u = x_f$, every solution has the displayed form, and every displayed vector clearly satisfies the equation. □

The inserted $-1$ rows in the practical trick are a bookkeeping device for the free variables. Choosing $u = e_j$, the $j$-th standard basis vector of the free-variable space, gives one kernel basis vector:

$$\begin{pmatrix} -Ae_j \\ e_j \end{pmatrix}.$$

So the trick is best read as a way of writing the graph basis of a kernel.

**Remark 1.13.2** (Do not confuse two uses of extra coordinates) — Homogeneous coordinates do turn affine transformations into linear transformations one dimension higher. That is a real and useful idea. But the minus-one trick in this row-reduction context is different: it does not linearize an affine map; it packages the free-variable choices in a homogeneous linear system.

## §1.14 Kernel-image factorization

For a linear map $T : V \to W$, the kernel records lost directions and the image records reachable directions. Rank-nullity says

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

## §1.15 Coordinates, kernels, and affine linearization

The review should distinguish vectors from coordinates, affine maps from linear maps, and a matrix from the linear transformation it represents. Kernel, image, and basis change are the structural core.

## §1.16 Exact sequences behind rank-nullity

For a linear map $T : V \to W$, the kernel and image fit into the exact sequence

$$0 \to \ker T \to V \xrightarrow{T} \operatorname{im} T \to 0.$$

Exactness means that the image of each arrow equals the kernel of the next. Taking dimensions gives

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

This is rank-nullity.

The elementary theorem is therefore the dimension shadow of an exact sequence. Exact sequences become one of the central languages of algebra, topology, and geometry.

## §1.17 Quotient description of the image

The first isomorphism theorem says

$$V/\ker T \cong \operatorname{im} T.$$

The quotient collapses all vectors that differ by a null vector. What remains is precisely the information visible after applying $T$.

This gives a clean meaning to free variables and solution spaces. The kernel is what the map forgets; the image is what survives.

## §1.18 Cokernel and obstructions

The cokernel of $T : V \to W$ is

$$\operatorname{coker} T = W/\operatorname{im} T.$$

It measures the failure of $T$ to be onto. A system

$$Tx = b$$

is solvable exactly when $b$ maps to zero in the cokernel.

Thus kernel measures non-uniqueness, while cokernel measures obstruction to existence. This is the conceptual form of the elementary distinction between infinitely many solutions and no solution.

## §1.19 Kernel, image, and quotient survive basis changes

The minus-one trick, basis conversion, and kernel/image computations in the note are all examples of one structure: a linear map forgets some directions, preserves some information, and may fail to reach some outputs. Exact sequences and quotients express this without depending on a chosen matrix.

## §1.20 Row reduction as choosing a splitting

When a matrix is row-reduced, the pivot variables are expressed in terms of free variables. Abstractly, this chooses a splitting of the domain into a complement of the kernel plus the kernel:

$$V \cong U \oplus \ker T.$$

The map $T$ is injective on the chosen complement $U$ and maps $U$ isomorphically onto $\operatorname{im} T$.

This splitting is not canonical: different row-reduction choices or different bases may choose different complements. The exact sequence

$$0 \to \ker T \to V \to \operatorname{im} T \to 0$$

is canonical; a particular parametric solution is a coordinate choice.

## §1.21 Cokernel as the missing equation space

For $T : V \to W$, the cokernel $W/\operatorname{im} T$ measures which right-hand sides are not reachable. In matrix terms, left null vectors produce compatibility conditions. If $y \in \ker T^T$, then any solvable equation $Tx = b$ must satisfy

$$y^T b = 0.$$

Thus the left null space is the dual of the obstruction space.

This gives a precise interpretation of consistency conditions in linear systems: they are not extra tricks, but equations defining whether $b$ vanishes in the cokernel.